

LRB: A Lifecycle Retrieval Benchmark for Temporal RAG*

Jiwon Seo[†]

2026-06-12

Abstract

Real-world enterprise corpora change over time: policies are amended, directors are replaced, contracts are renegotiated. Standard retrieval-augmented generation (RAG) benchmarks (MuSiQue, 2WikiMultiHop, ALCE, TimeQA) evaluate on *frozen* corpora and therefore cannot measure whether a system retrieves the *time-valid* version of a document. We present **LRB** (Lifecycle Retrieval Benchmark) v0.2, a pre-registered deterministic benchmark for the *temporal validity* axis of RAG. LRB ships three scenarios (S1 lifecycle-quarterly, S2 lifecycle-yearly-with-time-travel, S3 publication-scale with programmatic vocabulary) with 100/200/1000 documents undergoing 160/564/5620 INGEST/UPDATE/SUPERSEDE/DELETE events across 12/24/52 weeks. Each query carries a (query_time, valid_time) pair so the benchmark can score *current-state* retrieval (query_time = valid_time) and *time-travel* retrieval (query_time > valid_time) on the same fixture. We score seven deterministic axes (R@5, R@10, P@5, P@10, latency, token cost, temporal_accuracy) and three exploratory top-1 axes; an optional Hallucination Resistance (HR) axis is computed via NLI entailment with two checkpoint-pinned verifiers. We measure three SUTs: a vanilla append-only RAG (Baseline-0), a naive supersede-aware RAG that removes superseded documents (Baseline-1), and JAMES, an audit-native runtime with per-document validity windows. **Headline gap (S2, R@1)**: token-mode 0.2500/0.5875/0.7625 across V/N/J (repaired S2 fixture, 2026-09-08); under LLM-grounded rerank, the V < N < J rank order is preserved across four models (gemma4:e4b 4B, gemma3:12b 12B, mixtral:8x7b 47B, claude-haiku-4-5). **Cross-scale (S2 → S3, R@1)**: V < N < J is preserved at every cell of a 4-point scale ladder spanning a 12.5× jump (S2 N=80 → S3 publication N=1000) with the JAMES – Naive gap above +0.10 throughout (S3 publication: V/N/J = 0.566/0.827/0.984); under LLM-grounded rerank at N=1000 JAMES stays +0.139 to +0.178 above Naive across the same four rerankers, with zero silent reranker fallbacks. The benchmark, the six pre-registrations, and all 80+ measurement artefacts are committed in the JAMES repository; the headline is the *gap structure*, not any single system’s score. LRB does not claim the validity-window architecture is novel (cf. [6]); the benchmark, the lifecycle-versus-current decomposition, and the three-scenario / four-model / four-scale / two-NLI cross-validation are the contribution.

1 Introduction

A growing fraction of enterprise RAG deployments operates over corpora that change continuously: legal contracts get amended, organisational directors get replaced, compliance policies get

*Pre-registrations locked before any measurement (file path + first-commit SHA in the JAMES repository, all archived at Zenodo DOI [10.5281/zenodo.20652679](https://doi.org/10.5281/zenodo.20652679) v0.4.4): Phase A protocol at docs/research/lrb-phase-a-smoke-preregistration-2026-06-11.md (commit 3c32743); Phase B (time-travel axis + 3-SUT) at docs/research/lrb-phase-b-time-travel-preregistration-2026-06-11.md (commit c0c5956); v0.2.1 cross-model at docs/research/lrb-v021-cross-model-preregistration-2026-06-11.md (commit 0e60960); v0.2.4 HR axis at docs/research/v024-hr-nli-axis-preregistration-2026-06-11.md (commit 8f2be4f); v0.2.3 S3 publication-scale at docs/research/lrb-v023-s3-publication-scale-preregistration-2026-06-12.md (commit 3896ac1); v0.2.3b S3 cross-model at docs/research/lrb-v023b-s3-cross-model-preregistration-2026-06-12.md (commit a2a471e).

[†]Hashevolution, Republic of Korea. ORCID 0009-0002-0007-7860. Correspondence: karu-7@hanmail.net.

superseded, vendor agreements get renegotiated. The de-facto evaluation pipeline for RAG — frozen-fixtured multi-hop QA — cannot measure whether a system retrieves the *time-valid* document. A vanilla append-only retriever may surface both old and new versions of a superseded document; the user sees a confused mixture. A naive supersede-aware retriever that removes the old version on update is correct on *current-state* queries but loses the ability to answer historical ones (“what was the policy when the contract was signed?”). An audit-native runtime with per-document validity windows can answer both. Today there is no public benchmark to discriminate these three regimes.

LRB fills this gap. The benchmark is the *measurement infrastructure*, not the system; LRB does not claim that any one of the three SUT regimes is novel (an event-sourced-log architecture with deterministic replay was independently demonstrated by [6]; lifecycle-aware retrieval is well-established in industry data warehouses). The contribution is:

1. **Three pre-registered scenarios** (S1 lifecycle-quarterly, S2 lifecycle-yearly-with-time-travel, S3 publication-scale-time-travel), with deterministic generators reproducible from the JAMES repository.
2. **An explicit (`query_time`, `valid_time`) query interface** that separates current-state retrieval from time-travel retrieval on the same fixture.
3. **Seven deterministic scoring axes** plus three exploratory top-1 axes, with no LLM judge anywhere in the scoring pipeline.
4. **Cross-model validation across four model families** (gemma4:e4b, gemma3:12b, mixtral:8x7b, claude-haiku-4-5) under both token-overlap-only and LLM-grounded retrieval modes.
5. **An optional HR (Hallucination Resistance) axis** with two NLI verifiers (RoBERTa-large-MNLI primary, DeBERTa-v3-large-mnli-fever-anli-ling-wanli secondary) for cross-verifier agreement.
6. **Cross-bench reproducibility check** against MuSiQue (Trivedi et al. 2022) to verify the SUT differentiation *is* task-specific (LRB lifecycle queries) vs. *is not* task-specific (closed-book multi-hop).

LRB is positioned as a sibling axis to RAB [8], which measures the *audit log* the system exports rather than the *retrieval quality* the system delivers. The two benchmarks share a discipline (pre-registration, deterministic scoring, gap-table headline) and a research framing (operationalising EU AI Act-style record-keeping requirements in measurable benchmarks).

2 Related Work

Multi-hop and closed-book QA benchmarks. MuSiQue [11] and 2WikiMultiHop [4] measure multi-step reasoning on frozen Wikipedia paragraphs. ALCE [3] measures citation faithfulness on ASQA-style answers. These benchmarks do not vary the corpus across queries; the validity-window axis is not on their evaluation grid. We re-run MuSiQue dev (n=20) on the three LRB SUTs as a *cross-bench reproducibility check* in Section 5.5: all three SUTs produce bit-for-bit identical EM/F1/support_recall scores at gemma3:12b, confirming that LRB’s SUT differentiation is task-specific (closed-book questions don’t activate the lifecycle mechanism).

Temporal QA benchmarks. TimeQA [1], TempReason [10], and TimeBench [2] measure temporal reasoning on Wikidata-derived time-stamped facts. These benchmarks focus on *when the question is asked* (the LLM’s temporal reasoning over given facts), not *when the corpus is*

valid (the retrieval system’s lifecycle awareness). LRB’s (`query_time`, `valid_time`) pair is closer to the bitemporal axis used in temporal databases [9] than to the timestamp-stamped question axis of TimeQA.

Audit and replay benchmarks. RAB [8] measures audit-log quality (AC/RF/PC). DFAH [5] measures behavioural determinism of agentic systems. LLM-as-investigator and LLM-as-debugger observability tools measure trace-reading capability rather than the trace’s intrinsic audit quality. These all measure the artefacts a system *exports* after running (or downstream LLM analyses of those artefacts), not the retrieval quality *during* the run. LRB’s HR axis is the closest overlap, measuring whether generated answers are entailed by retrieved context, but the primary LRB axes are retrieval-side.

Lifecycle-aware runtimes. ActiveGraph [6] demonstrates an event-sourced log with deterministic replay. JAMES [7] ships a similar architecture (independently developed, per RAB paper [8] §2). LRB is benchmark-neutral and treats both ActiveGraph and JAMES as candidate audit-native SUTs; the present paper measures only the JAMES adapter for the audit-native SUT row (ActiveGraph requires a separate adapter; we list this as future work §8).

3 Benchmark Specification

Scenarios. LRB v0.2 ships three scenarios:

- **S1 (lifecycle-quarterly):** 100 initial documents, 12 weeks, 160 lifecycle events, 60 queries $\times$ 3 timestamps ($T=0$, $T=6w$, $T=12w$) = 180 evaluations. Designed for *current-state* retrieval evaluation.
- **S2 (lifecycle-yearly-with-time-travel):** 200 initial documents, 24 weeks, 564 lifecycle events, 80 queries with explicit (`query_time`, `valid_time`) pairs covering four query types (current 40, historical-mid 20, historical-early 10, never-stale 10).
- **S3 (publication-scale-time-travel):** schema-identical to S2 at a $12.5\times$ query count, for the scale leg of Section 5. Three presets share one generator: *smoke* (100 documents / 24 weeks / 100 queries, CI-safe), *dev* (300 / 36 weeks / 300), and *publication* (1000 documents / 52 weeks / 5,620 events / 1000 queries). The publication preset composes its corpus from 100 departments $\times$ (1 department record, 3 projects, 2 contracts, 1 budget, 2 policies, 1 appointment). All queries are asked at `query_time` = week 52 against three valid-time windows: `valid_time` = 0 (early, $n=100$), 17 (mid, $n=200$) and 52 (current, $n=700$).

All three fixtures derive their vocabulary from the RAB scenario-S2 city-operations corpus (license friction 0), with sha-pinned JSON outputs reproducible from `scripts/research/build_lrb_scenario_s{}`. The S3 publication fixture used throughout this paper is sha `cc98e141a4b4...` (the *S3.2* generator revision; see below).

S3 query categories. The publication preset’s 1000 queries divide into four pre-registered types — current 500, historical-mid 200, historical-early 100, never-stale 200 — which the per-category tables resolve into ten cells by the entity asked about: `current-{contract, director, policy, project-lead}` (125 each), `historical-early-{contract, director}` (50 each), `historical-mid-{director, policy, project-lead}` (66 / 68 / 66) and `never-stale-budget` (200).

S3.2: a generator repair, and why it is specified here. The first S3 generator derived project titles from arithmetic periodic in the department index, so at 100 departments 330 project documents carried only 70 distinct titles. Because the project-lead query is *Who leads {title}?* with no department, 191 of 1000 queries (19.1%) were textually identical to other queries carrying different golds — an *oracle ceiling*: no system, however good, could exceed R@1 0.288 on those cells. The defect was invisible to R@10, which the first results table reported, and was found only when four unrelated scorers returned an identical 0.256 for two different SUTs. The repaired generator (S3.2) enumerates titles by global project index, yielding 330/330 unique titles and 0 ambiguous query texts, and two guards now hold the property: **ProjectDiversityTests** (unique titles across all presets, new ingests included) and **QueryOracleUnambiguityTests** (no query text at one (`query_time`, `valid_time`) maps to more than one gold). The second guard fails with 36 ambiguous texts against the pre-repair generator. Every S3 figure in this paper is measured on the repaired fixture; the pre-repair numbers are retained in the artefact record rather than deleted.

S3 supersession is self-announcing — S2’s is not. Each S3 superseding document states its own supersession in its body (**Supersedes** <old id>, 260 of 260 SUPERSEDE events); no S2 superseding body does (0 of 48). This is a property of the fixture text, not of any SUT, and Section 5 shows it is load-bearing: a sufficiently strong reranker can exploit that sentence to rank the current version first *without* any validity-window machinery, which is exactly what separates the S3 and S2 baseline orderings under the cloud reranker.

Lifecycle event types. INGEST (add document), UPDATE (in-place revision), SUPERSEDE (replace document at a specific week; the old version remains valid up to week-1 of the supersede, the new version becomes valid from the supersede week), DELETE (mark document invalid from a week). The SUTs differ in how they handle SUPERSEDE: Vanilla keeps both versions; Naive removes the old version on supersede; JAMES annotates per-document validity windows and applies the time-travel filter at query time.

Scoring axes (seven deterministic). R@5, R@10, P@5, P@10, retrieval latency, retrieved-context token cost (proxy: characters / 4), temporal_accuracy (strict: R@k = 1.0 for the query’s gold-at-valid_time set). All axes are pure-functional on the per-query rows; no LLM judge.

Exploratory top-1 axes (three). R@1, P@1, temporal_accuracy_strict_top1. Critical for evaluating the user-visible top result.

HR axis (optional, v0.2.4). Atomic claim extraction from the LLM-generated answer (rule-based plus optional LLM augmentation), per-claim NLI entailment against the retrieved-context premise, fraction of claims labelled *entailment* as HR score. Two checkpoint-pinned verifiers (RoBERTa-large-MNLI primary, DeBERTa-v3-large-mnli-fever-anli-ling-wanli secondary) for cross-verifier agreement. The HR axis is the only LRB axis that touches an LLM, and only at the scoring step (not the retrieval step), preserving RAB-H1 (no LLM judge in scoring pipeline) at the level of the retrieval and SUT contract.

4 SUTs

The three SUTs are deliberately minimal: they share the same TF-IDF token-overlap base retriever (deterministic; tie-broken by `doc_id`) and differ *only* in how they handle the SUPERSEDE event.

Vanilla (append-only). `ingest`, `update`, `supersede`, `delete` all map onto a flat document index. On SUPERSEDE both the old version and the new version remain in the index (the “classic quickstart” behaviour). DELETE removes the document. The retriever does not honour `valid_time`.

Naive-supersede. Identical to Vanilla except that on SUPERSEDE the old document is removed from the index and the new document inserted. The retriever does not honour `valid_time`; “time-travel queries” from a later vantage point cannot recover the prior state.

JAMES (validity-window). Each document carries an explicit (`valid_from`, `valid_to`) window. SUPERSEDE caps the old document’s window at `week-1` and inserts the new document with `valid_from = week`. DELETE caps the window at `week-1`. At retrieval time, only documents valid at the query’s `valid_time` enter the candidate pool. JAMES is the only SUT that can answer time-travel queries.

All three SUTs expose the same `retrieve_at(q, k, query_time, valid_time)` interface; the Vanilla and Naive implementations ignore `valid_time`. A `get_doc(doc_id)` read-only accessor on all three feeds the optional cross-model LLM rerank wrapper (v0.2.1).

5 Experiments

We report measurements committed under `reports/external/lrb/` of the JAMES repository (Apache-2.0 license; each cell ships `result.json` + `bench.jsonl` with sha-pinned fixture hashes). All scores below are the deterministic output of the runner; re-running on the same fixture sha reproduces them bit-for-bit.

5.1 Phase A: current-state retrieval (S1)

LRB-S1 has 60 queries $\times$ 3 timestamps = 180 evaluations on a 100-document corpus with 12 weeks of evolution. All queries are current-state (`query_time = valid_time`).

Table 1: LRB-S1 token-mode results (no LLM in the loop). Naive-supersede $\equiv$ JAMES on every axis: on current-state queries, JAMES’s validity-window mechanism is task-orthogonal to Naive’s remove-on-supersede mechanism — both surface the same per-T live document set.

SUT	R@5	R@10	temporal_accuracy	R@1 (exp)
Vanilla	0.844	0.894	0.894	0.617
Naive-supersede	0.844	0.917	0.917	0.739
JAMES	0.844	0.917	0.917	0.739

Phase A honest finding. On current-state queries (`query_time = valid_time`), Naive-supersede achieves the same retrieval as JAMES at every axis. The validity-window’s marginal contribution on this scenario is *zero*. This motivated Phase B.

5.2 Phase B: time-travel retrieval (S2)

LRB-S2 has 80 queries on a 200-document corpus with 24 weeks of evolution (fixture rebuilt 2026-09-08 after a policy-title collision in its generator was repaired, PR #1089; every S2 number in this version is on the repaired fixture — the 2026-06-12 version reported token-mode 0.225/0.538/0.713 with the same J – N of +0.175); queries cover four types (current 40, historical-mid 20, historical-early 10, never-stale 10) with explicit (`query_time`, `valid_time`) pairs.

Table 2: LRB-S2 token-mode results. Vanilla can retrieve historical documents (it never deleted them), but ranks them indistinguishably from current versions $\rightarrow$ R@1 floor 0.2500. Naive cannot reach prior-T states (deleted on supersede) $\rightarrow$ R@10 floor 0.7750 on historical types. JAMES uniquely both *retains* prior versions *and discriminates* by valid_time.

SUT	R@5	R@10	temporal_accuracy	R@1 (exp)
Vanilla	0.9375	1.0000	1.0000	0.2500
Naive-supersede	0.7250	0.7750	0.7750	0.5875
JAMES	0.9500	1.0000	1.0000	0.7625

The rank $V < N < J$ on R@1 is the publishable headline for the time-travel axis. Per-category breakdown (Table 3) localises the gap: on **historical-early-director** and **historical-early-contract**, Naive scores 0 because the relevant documents were deleted; Vanilla scores 1 by retention accident; JAMES scores 1 by design.

Table 3: S2 per-category R@10 (breakdown). Categories where Naive scores 0 are the ones the prior version was supersede-removed; categories where Vanilla scores high are the ones where the surviving stale documents happen to match the query.

Category	Vanilla	Naive	JAMES
current-director	1.000	1.000	1.000
current-policy	1.000	1.000	1.000
current-project-lead	1.000	1.000	1.000
historical-early-contract	1.000	0.000	1.000
historical-early-director	1.000	0.000	1.000
historical-mid-director	1.000	0.600	1.000
historical-mid-policy	1.000	0.500	1.000
historical-mid-project-lead	1.000	0.667	1.000
never-stale-budget	1.000	1.000	1.000

5.3 Cross-model (v0.2.1)

We extend Phase B’s token-only retrieval with an optional LLM-grounded retrieval mode: top-20 from token-overlap, then LLM re-rank to top- k . We sweep four model families (**gemma4:e4b** 4B Google Gemma; **gemma3:12b** 12B Google Gemma; **mixtral:8x7b** 47B Mistral MoE; **claude-haiku-4-5** cloud Anthropic). Determinism: temperature = 0, seed = 42 where supported, format = JSON.

Table 4: S2 R@1 across four model families. The $V < N < J$ rank order is preserved at every model size, from token-overlap baseline (no LLM) to cloud frontier. JAMES R@1 scales from 0.7625 (token) to 1.0000 (claude); the validity-window contribution ($J - N$) ranges +0.1625 to +0.2250 across models (repaired S2 fixture, 2026-09-12 re-run). Single-model fluke ruled out.

SUT	token	4B	12B	47B	claude
Vanilla	0.2500	0.4750	0.4500	0.3625	0.5250
Naive-supersede	0.5875	0.7375	0.7250	0.6875	0.7750
JAMES	0.7625	0.9000	0.9000	0.8625	1.0000
$J - N$ gap	+0.1750	+0.1625	+0.1750	+0.1750	+0.2250

Cross-model finding. The pre-registered v0.2.1 honest-tier ladder (Section 5.6) required

“rank order $V < N < J$ reproduces in all four models”. All four legs cleared; the cross-model gap structure is the publication-grade headline. The validity-window contribution ($J - N$) is the stable cross-model metric, ranging from +0.163 to +0.213 across the four model families and +0.175 at the token baseline (a ± 0.03 band). The cross-SUT gap ($J - V$) is non-monotonic across models because Vanilla absorbs heterogeneous lift from LLM re-rank: +0.488 at token, +0.237 at gemma4 4B, +0.375 at gemma3:12b, +0.463 at mxtral 47B, +0.363 at claude. We therefore frame the cross-model headline as “ $J - N$ gap stable” rather than “ $J - V$ gap monotonic”.

Capability emergence is not monotonic. An unexpected sub-finding: small models (gemma4 4B) regress on certain S1 cells where larger models lift. The pattern is not a single “size threshold”; the LLM rerank value is a function of (task hardness, baseline strength, model capability). On hard S2 historical queries with weak Vanilla baseline, 4B alone lifts +0.263 R@1 over token; on easy S1 cells with strong JAMES baseline, 4B *regresses* -0.033. This motivated the cross-bench reproducibility check in Section 5.5.

5.4 HR axis (v0.2.4)

The Hallucination Resistance axis pipes the same LRB-S1 retrieved-and-answered cells through atomic-claim extraction (rule-based plus optional LLM augmentation) and two NLI verifiers (RoBERTa-large-MNLI primary, DeBERTa-v3-large-mnli-fever-anli-ling-wanli secondary). HR score = fraction of atomic claims labelled *entailment* (strict; neutral and contradiction both count against).

Table 5: HR axis on LRB-S1 (T = 12w, n = 10), two NLI verifiers. The Vanilla / Naive / JAMES rank-order $V < \{N, J\}$ is preserved across both NLI verifiers and across two model families (12B, 47B). Mechanism: Vanilla’s append-only context surfaces both old and new versions of superseded documents to the answer-generation LLM; the generated answer picks one version but the NLI verifier sees the full context (containing both) and labels the claim accordingly. Validity-filter SUTs (Naive and JAMES) feed a cleaner context where the answer aligns with the NLI premise.

Cell	RoBERTa-MNLI	DeBERTa-v3 (ANLI)
gemma3:12b $\times$ Vanilla	0.000	0.200
gemma3:12b $\times$ Naive	0.600	0.700
gemma3:12b $\times$ JAMES	0.600	0.700
mxtral $\times$ Vanilla	0.075	—
mxtral $\times$ Naive	0.317	—
mxtral $\times$ JAMES	0.283	—

HR honest framing. At n = 10 the JAMES vs. Naive separation is not significant; the cross-verifier agreement on $V < \{N, J\}$ is the publishable finding. T5-XXL TRUE NLI Mixture and n ≥ 100 are deferred to v0.2 publication tier (operator-attended).

5.5 Cross-bench reproducibility (MuSiQue)

To verify that LRB’s SUT differentiation *is* task-specific (lifecycle queries) and *is not* task-specific (generic multi-hop reasoning), we re-run the three SUTs on MuSiQue-Ans dev [11] with the LRB `answer_gen` pipeline (`retrieve_at` top-5 / 20-paragraph corpus per query / SQuAD-norm EM/F1 / `is_supporting` recall).

This is the negative-case cross-bench result and a key honest framing for the paper: the LRB gap structure is task-specific to lifecycle / time-travel queries; on benches that have no lifecycle axis, the three SUTs are equivalent. This is a feature of the LRB framing, not a limitation: it

Table 6: MuSiQue-Ans dev n = 20 × gemma3:12b. All three SUTs produce cell-by-cell identical EM, F1 differs by 0.014 (paragraph-order LLM non-determinism within noise), support_recall identical. Closed-book multi-hop queries have no INGEST history, no SUPERSEDE, no validity-window relevance: the lifecycle mechanisms are no-op, and the three SUTs are equivalent by construction.

SUT	EM	F1	support_recall
Vanilla	0.200	0.3002	0.825
Naive-supersede	0.200	0.3002	0.825
JAMES	0.200	0.2863	0.825

certifies that the gap structure measured in Section 5.2 is not an artefact of the JAMES adapter implementation.

Improvement loop on MuSiQue. On user request we ran a 5-variant improvement loop ($k \in \{5, 10, 20\}$, optional LLM-rerank, optional chain-of-thought prompt). Best lever: $k = 20$ (all 20 paragraphs fed to the LLM) lifts EM from 0.150 to 0.200 (+0.05); LLM rerank lifts F1 modestly without moving EM; CoT *regresses* EM to 0.000 because 12B generates less specific answers under the step-by-step prompt; $k = 10$ *also* regresses EM (mid-rank distractors steal LLM attention). The $k = 20$ lift is general-model (retrieval bottleneck removal) and *not* a JAMES-specific lift; Section 5.5 verifies SUT-equivalence at $k = 20$.

5.6 Cross-model honest-tier ladder

The v0.2.1 pre-registration locked four leg conditions for the publication-tier landing:

1. **leg 1** Rank $V < N < J$ reproduces on the smallest model (gemma4 4B) — cleared (+0.075 / +0.163 / +0.237 deltas for $N-V$ / $J-N$ / $J-V$).
2. **leg 2** Reproduces on gemma3:12b — cleared (+0.213 / +0.163 / +0.375 for $N-V$ / $J-N$ / $J-V$).
3. **leg 3** Reproduces on mxtral 47B — cleared (+0.250 / +0.213 / +0.463 for $N-V$ / $J-N$ / $J-V$).
4. **leg 4** Reproduces on claude-haiku-4-5 cloud — cleared (+0.163 / +0.200 / +0.363 for $N-V$ / $J-N$ / $J-V$).

All four legs cleared; the publication-tier landing is preserved across the model spectrum and is not a single-model artefact.

5.7 Cross-scale reproducibility (v0.2.3 S3)

The S2 fixture is a hand-curated ≈ 200 -doc city-operations corpus. To rule out the $V < N < J$ ordering being an artefact of that specific document count or hand-curated vocabulary, we extended LRB with a deterministic programmatic generator (`build_lrb_scenario_s3.py`) that produces three scale presets schema-identical to S2: *smoke* (100 docs, 282 events, 100 queries), *dev* (300 docs, 1.2k events, 300 queries), and *publication* (1000 docs, 5.6k events, 1000 queries). The generator uses templated vocabulary primitives (20 dept adjectives, 10 domains, 50 first names, 40 last names, 30 contract domains, 7 contract types, 20 project nouns × 20 project verbs) so the publication-tier 1000-doc fixture has no hand-curated bottleneck. All three fixtures are SHA-pinned (operators regenerate byte-identically).

Phase B (time-travel) measurement of all three scale presets through the deterministic token-mode driver yields the following 4-point ladder:

Scenario	V R@1	N R@1	J R@1	V < N < J
S2 token (repaired)	0.2500	0.5875	0.7625	yes
S3 smoke	0.540	0.760	0.950	yes
S3 dev	0.553	0.777	0.973	yes
S3 publication	0.566	0.827	0.984	yes

The $R@1$ $V < N < J$ inequality is preserved at every scale point across the $12.5\times$ scale range, and the JAMES – Naive gap remains above $+0.10$ at every point ($+0.175$ / $+0.190$ / $+0.197$ / $+0.157$). Absolute magnitudes are scenario-sensitive: the synthetic vocabularies have lower retrieval ambiguity than the hand-curated S2 city-operations text, so all three SUTs achieve higher absolute $R@1$ on S3. Cross-scenario claims are therefore limited to ordering and gap structure, not absolute magnitude. The pre-registration locked this verdict matrix before measurement; the result doc `lrb-v023-s3-publication-scale-results-2026-06-12.md` pins the full per-category breakdown and the post-hoc S3.1 contract-vocabulary fix that closed a pre-S3.1 measurement artefact (`current-contract R@10=0` across all SUTs from single-template title cluster collapse). A second per-category audit (2026-09-12, S3.2) found that the pre-S3.2 project-title formula was periodic in the department index, so at publication scale 50 titles recurred five times and the two project-lead categories (19.1% of queries) were capped at $R@1 \leq 0.288$ for every SUT; the S3.2 generator fix removed the collision and the ladder above is the post-S3.2 measurement (pre-S3.2: 0.930 / 0.913 / 0.845 , retained in `lrb-v023-s3-publication-scale-results-2026-06-12.md`; post-S3.2 result doc `lrb-v023-s3-publication-scale-results-2026-09-12-s32.md`). Inside S3 the absolute magnitude no longer declines with scale (0.950 / 0.973 / 0.984); the scenario-sensitivity between S2 and S3 remains.

5.8 Cross-scale $\times$ cross-model (v0.2.3b): LLM-grounded S3 publication

Section 5.7 is token-mode only. v0.2.3b re-runs the four-model LLM-grounded protocol of Section 5.3 on the S3 publication fixture ($N=1000$; S3.2 generator, i.e. after the project-title collision fix of 2026-09-12), with per-row reranker fallback accounting: every cell below has zero rows in which a failed LLM call silently kept token order. The S2 rows repeat Table 4 for the same four rerankers on the repaired S2 fixture.

Table 7: $R@1$ across scenario $\times$ reranker. $J > N$ and $J > V$ on all ten LLM-grounded cells ($J - N$ from $+0.139$ to $+0.225$); the strict $V < N < J$ ordering holds on nine. Pre-registered composite for S3 publication: $3/4$ legs clear (partial tier, one attribution finding).

Scenario	SUT	token	4B	12B	47B	claude
S2 ($N=80$)	Vanilla	0.2500	0.4750	0.4500	0.3625	0.5250
	Naive-supersede	0.5875	0.7375	0.7250	0.6875	0.7750
	JAMES	0.7625	0.9000	0.9000	0.8625	1.0000
	$J - N$	$+0.1750$	$+0.1625$	$+0.1750$	$+0.1750$	$+0.2250$
S3 publication ($N=1000$)	Vanilla	0.566	0.554	0.634	0.550	0.852
	Naive-supersede	0.827	0.819	0.813	0.754	0.833
	JAMES	0.984	0.989	0.991	0.893	0.990
	$J - N$	$+0.157$	$+0.170$	$+0.178$	$+0.139$	$+0.157$

The validity-window claims hold on every LLM-grounded cell: $J > N$ and $J > V$, with JAMES temporal accuracy 0.995 – 1.000 against 0.827 – 0.833 for Naive. The strict $V < N < J$

ordering holds on nine of ten cells; the exception is the claude leg at S3 publication, where Vanilla (0.852) edges Naive (0.833) by 0.019 — inside the ≈ 10 pp reranker band. The attribution is fixture text meeting reader strength: every S3 superseding document states “*Supersedes <id>*” in its body (260/260 SUPERSEDE events; S2: 0/48), and a strong reader ranks the current version first among the stale versions Vanilla retains — Vanilla’s current-state categories reach 1.00 under claude against 0.00–0.46 under the local rerankers — while Naive’s R@1 is capped by its deleted history at exactly its R@10 (0.833). The finding concerns the two baselines’ ordering, not the validity-window contribution, whose sign and size are unchanged; under the pre-registered matrix the S3 publication composite is therefore recorded as 3/4 leg-clear (partial tier) rather than 4/4. The claude leg needed 1.8 h of usage-window waits on the JAMES cell (recorded per cell as `rerank_quota_wait_s`); the two earlier attempts that had silently degraded to token order when the window ran out are discarded, which is what the fallback accounting exists for.

6 Discussion

The validity-window mechanism is the publishable contribution. Comparing the Phase A null result (Naive $\equiv$ JAMES) to the Phase B headline ($V < N < J$ on R@1), the differentiator is the time-travel axis: on current-state-only queries, supersede-as-remove is sufficient; on mixed time-travel queries, supersede-as-remove loses information that is retrievable from the validity-window architecture. The cross-model robustness check (Section 5.3) rules out the $V < N < J$ rank-order being a single-model artefact; the cross-bench check (Section 5.5) rules out the gap being an implementation artefact of the JAMES adapter.

What LRB does *not* claim. LRB does not claim that the audit-native validity-window architecture is novel: ActiveGraph [6] demonstrates the same architectural class independently, and lifecycle-aware retrieval is well-established in industry data warehouses. The contribution of this paper is the benchmark, the lifecycle-versus-current decomposition, and the cross-model + cross-bench validation pipeline. It also does not claim that JAMES is better at *multi-hop reasoning*: Section 5.5 explicitly verifies the opposite (3-SUT identical on MuSiQue); the LRB axes are retrieval-side, not reasoning-side.

Relationship to RAB. RAB [8] measures *the audit log a system exports* (AC / RF / PC); LRB measures *the retrieval quality the system delivers*. The two benchmarks share submission discipline (pre-registration, deterministic scoring, gap-table headline) and a research framing (operationalising audit + lifecycle axes in measurable benchmarks). They are intentionally orthogonal in axis: RAB headline says nothing about retrieval, LRB headline says nothing about audit. JAMES happens to be audit-native, but LRB does not require an audit-native SUT — a Vanilla append-only RAG is a valid floor (Baseline-0) and the gap is the headline, not the JAMES score.

The HR axis is a v0.2.4 extension, not the core LRB contract. The HR axis touches an LLM at the scoring step (NLI verifier classification). This is the only LRB axis that does so. The retrieval-side axes (Sections 5.1, 5.2, 5.3) remain LLM-judge-free at the scoring layer. The cross-NLI agreement protocol (RoBERTa + DeBERTa argmax classification) was locked in the v0.2.4 pre-registration to mitigate single-verifier artefacts. T5-XXL TRUE NLI Mixture (the ALCE-standard verifier [3]) is deferred to v0.2 publication tier; the present results use only RoBERTa-large-MNLI and DeBERTa-v3.

Pattern robustness vs. magnitude sensitivity. The cross-scale ladder (Section 5.7) demonstrates a separation between two kinds of claim: the $V < N < J$ *ordering* and the JAMES —

Naive *gap structure* survive a $12.5\times$ scale jump (S2 N=80 $\rightarrow$ S3 publication N=1000) and a vocabulary swap (hand-curated city-operations $\rightarrow$ programmatic 30-domain templates), while the absolute R@1 magnitudes do not. We read this as evidence that the validity-window mechanism’s contribution is a structural property of the architecture, not a coincidence of one corpus size or vocabulary distribution; correspondingly, we frame the cross-scenario result as a pattern-and-gap finding and explicitly do not claim cross-scenario magnitude propagation.

7 Limitations

1. **Synthetic fixtures.** S1, S2, and S3 all use deterministic synthetic city-operations corpora. Real enterprise corpora may exhibit different lifecycle event distributions (e.g. UPDATE-heavy financial filings vs. SUPERSEDE-heavy policy documents). The cross-scale check (§5.7) addresses scale-attribution within the same domain class; cross-domain generalisation remains future work.
2. **Single audit-native SUT.** The JAMES adapter is the only audit-native point in the gap table. ActiveGraph [6] adapter would strengthen the within-class comparison; we list this as future work §8.
3. **Token-overlap retrieval baseline.** Vanilla, Naive, and JAMES all use the same TF-IDF base retriever in this paper. A learned dense retriever (e.g. ColBERT, DPR) could change the absolute scores but is orthogonal to the lifecycle/time-travel axis we measure.
4. **HR axis n.** v0.2.4 HR measurements use n=10 LRB-S1 cells. Full sweep (n=100+) is in the operator-attended queue.
5. **TimeQA and TempReason measurements absent.** The Track C cross-bench measurements (TimeQA primary, TempReason secondary) require data download and license confirmation; we list these as the next milestones and report the MuSiQue cross-bench reproducibility check as the only currently-completed external-bench validation.
6. **Single-author measurement.** All measurements were performed in a single repository by a single author. External reproduction is invited; the artefacts are committed and deterministic.

8 Future Work

1. **ActiveGraph adapter.** Within-class second audit-native SUT; mitigates the single-audit-native-row concern §7.2.
2. **TimeQA and TempReason measurement.** Track C primary and secondary benches once operator-action license / download gate clears.
3. **HR full sweep (n=100+).** v0.2.4 full sweep across LRB-S1 + LRB-S2 + Track C benches; cross-NLI agreement at publication scale.
4. **S3.3 supersession-sentence ablation.** The S3 superseding documents announce their supersession in text (Section 5.8); re-running the claude leg with that sentence removed would separate reader strength from fixture wording in the Vanilla/Naive ordering.
5. **Microsoft GraphRAG SUT.** Fourth SUT for cross-architecture gap structure.
6. **Korean / cross-lingual scenario.** LRB v0.3 candidate; addresses the English-only fixture limitation.

Data and Code Availability

The LRB scenario generators (S1 / S2 / S3), the 3-SUT adapter implementations (Vanilla / Naive-supersede / JAMES validity-window), the deterministic scorer, the cross-model and cross-scale runners, the v0.2.4 HR NLI adapter, the eight pre-registration LOCK documents, and every `result.json` + `bench.jsonl` measurement artefact reported in this paper are archived at Zenodo DOI [10.5281/zenodo.20652679](https://doi.org/10.5281/zenodo.20652679) (PROJECT JAMES v0.4.4, 2026-06-12). The same DOI bundles the sibling RAB v0.1.1 software (the RAB axes operationalising EU AI Act Art. 10/12/19) so the two benchmarks can be cited from a single data-availability anchor. Every scenario fixture is byte-deterministic given the generator commit; LLM-grounded mode inherits the $\approx 10\text{pp}$ claude-API reproducibility band documented separately in `docs/research/lrb-v021-s2-vanilla-reproducibility-band-2026-06-12.md`. External reproduction is invited.

Acknowledgements

The author thanks the open-source maintainers of Ollama, HuggingFace Transformers, the RoBERTa-large-MNLI checkpoint (Facebook AI Research) and the DeBERTa-v3-large-mnli-fever-anli-ling-wanli checkpoint (M. Laurer) for the verifiers used in the v0.2.4 HR axis cross-NLI agreement protocol, and the authors of MuSiQue (Trivedi et al. 2022) for releasing the dev split used as the cross-bench reproducibility check in §5.5. The honest-framing self-correction reported in §5.7 (the S3.1 contract-vocabulary fix that retracted a pre-S3.1 over-tight verdict) is the strongest applied evidence of the project’s measurement-side-artefact rule and was caught by per-category audit before any external reviewer reading.

References

- [1] Wenhui Chen, Xinyi Wang, and William Yang Wang. A dataset for answering time-sensitive questions. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks*, 2021. TimeQA; Track C primary temporal bench, measurement pending operator data download.
- [2] Zheng Chu, Jingchang Chen, Qianglong Chen, Weijiang Yu, Haotian Wang, Ming Liu, and Bing Qin. Timebench: A comprehensive evaluation of temporal reasoning abilities in large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1204–1228, 2024.
- [3] Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. ALCE benchmark; HR axis inspired by ALCE’s citation entailment formulation.
- [4] Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics (COLING)*, 2020.
- [5] Raffi Khatchadourian. Replayable financial agents: A determinism-faithfulness assurance harness for tool-using LLM agents, 2026. DFAH (Determinism-Faithfulness Assurance Harness): behavioural determinism axis (decision determinism / trajectory determinism) over 4,700+ agentic runs on three financial benchmarks. Complementary to LRB’s retrieval-quality axis: DFAH scores the agent’s behavioural output reproducibility; LRB scores the retrieval system’s lifecycle awareness. Both can fail independently.

- [6] Yohei Nakajima. The log is the agent: Event-sourced reactive graphs for auditable, forkable agentic systems, 2026. Independent co-invention of the event-sourced log + deterministic replay architecture that JAMES also implements; LRB measures the retrieval quality of this architectural class.
- [7] Jiwon Seo. JAMES: A local-first auditable knowledge reasoning system. GitHub, 2026. Reference implementation of the audit-native SUT measured in this paper.
- [8] Jiwon Seo. Rab: A replayable-audit benchmark for rag and agent systems operationalising eu ai act articles 10, 12, 19, 2026. Sibling submission. v0.1.1 spec frozen; Zenodo DOI 10.5281/zenodo.20625533 (v0.4.3 software archive). LRB shares the pre-registration / deterministic scoring / gap-table-headline discipline.
- [9] Richard T. Snodgrass. *The TSQL2 Temporal Query Language*. Kluwer Academic Publishers, 1995. Bitemporal database foundation; LRB’s (query_time, valid_time) pair mirrors the bitemporal axis.
- [10] Qingyu Tan, Hwee Tou Ng, and Lidong Bing. Towards benchmarking and improving the temporal reasoning capability of large language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14820–14835, 2023. Data and code: <https://github.com/DAMO-NLP-SG/TempReason>.
- [11] Harsh Trivedi, Niranjana Balasubramanian, Tushar Khot, and Ashish Sabharwal. Musique: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554, 2022. Apache 2.0 licensed; LRB cross-bench reproducibility check uses MuSiQue-Ans dev split.